

NEET Important Chapter – Communication Systems

Communication Systems – An Important Topic of NEET

Search your topic here...

The term communication means the transferring, receiving and processing of information by electronic means. There are basically two communication modes : point to point and broadcast. In point to point mode of communication, the information is transferred from single transmitter to single receiver whereas in broadcast mode of communication, there are a large number of receivers corresponding to single transmitter.

In this chapter, students will learn about the basic communication system and different terms used in communication. Going further in this chapter, students will also learn about the meaning and need of modulation and demodulation.

In this article, we will cover the important concepts and topics that will help students to revise and boost their preparations for [NEET](#).

Important Topics of Communication Systems

- Basic Elements of Communication System
- Important Terms Used in Communication Systems
- Line Communication
- Optical Communication
- Optical Fiber
- Modulation
- Demodulation
- Propagation of Electromagnetic waves
- Satellite Communication

Important Concepts of Communication Systems

Sl.No	Name of the Concept	Key Points
1.	Basic Elements of a Communication Systems	• Transmitter converts the message signal produced by the information source into an electrical signal which can be carried through the channel to the receiver. • Communication channel carries the modulated wave from transmitter to receiver. • Receiver receives the signal from the communication channel and converts it into original form. (image will be uploaded soon)

2.	Important Terms Used in Communication systems	• Analog signal: Signal which varies Continuously w.r.t time. (image will be uploaded soon) • Digital signal: Signal which has two levels of voltage usually in the form of pulses, represented by 0 or 1. (image will be uploaded soon) • Transducer: It converts one form of energy into another. • Noise: It refers to the unwanted signal that tends to disturb the transmission. • Bandwidth refers to the range over which the frequencies in a signal vary. • Amplification is the process of increasing the amplitude and thus strength of electrical signal.
3.	Line Communication	• In line communication, there is a physical connection between source and the destination. The wired connections between two connections are known as transmission lines.
4.	Optical Communication	• Optical communication uses light waves in the frequency range 10^{12} to 10^{16} Hz as the guided wave medium for the propagation of audio frequency signals.

5.	Modulation	- The phenomenon of superposition of an information signal over a high frequency carrier wave is called modulation. - Digital and analog signals to be transmitted are usually of low frequency and hence, cannot be transmitted as such.
6.	Amplitude Modulation (AM)	- Amplitude Modulation is the process of changing the amplitude (A_c) of a carrier wave linearly in accordance with the amplitude of the message signal (A_m). - Modulation index (μ): —————
7.	Frequency Modulation(FM)	- Frequency Modulation is the process of changing the frequency of a carrier wave (ν_c) in accordance with the frequency of the message signal (ν_m). - Modulation index (μ_f) = — where, δ = maximum frequency of deviation
8.	Demodulation	The process of recovering the original audio signal from the modulated signal.

9.	Propagation of Electromagnetic Waves	• Ground Wave Propagation: In this type of communication, transmitting and receiving antennas are close to the surface of earth. • Sky Wave Propagation: Sky wave is the type of radio wave which is directed towards the sky and reflected back from the ionosphere towards the desired location of earth. • Space Wave Propagation: The transmitted signal is received by the direct interception of the signal by the receiving antenna. The maximum range of this transmission is given by, $d = \sqrt{2Rh}$ where, R = radius of earth and h = height of antenna
----	--------------------------------------	---

List of Important Formulae of Communication Systems

Sl. No	Name of the Concept	Formulae
1.	Modulation index (μ)	_____
2.	Voltage equation of Amplitude Modulated wave (e)	_____
3.	Modulation index (μ_f)	_____ where, δ = maximum frequency of deviation
4.	Height of Antenna	_____ If the height of receiving antenna is also given, then maximum line of sight distance (d_M) The minimum height of antenna (not of antenna tower) should be $\lambda/4$ where λ is wavelength of modulating signal.

Solved Examples of Communication Systems

1. A TV tower has a height of 60 m. What is the maximum distance and area up to which TV transmission can be received? (Take the radius of earth as _____ m.)

Sol:

Given,

Height of the tower (h) = 60 m,

Radius of the earth (R) = _____ m,

Maximum distance upto which TV transmission (d) can be received is given by:



Now, area (A) upto which transmission can be received will be:

$$A = \pi d^2 =$$

$$A = 2411 \text{ km}^2$$

Therefore, distance and area upto which transmission can be received is 27.7 km and 2411 km², respectively.

Key Point: If the height of receiving antenna is also given, then maximum line of sight distance (d_M),

2. What is the purpose of modulating a signal in transmission?

Sol:

A low frequency signal cannot be transmitted to long distances because of many practical difficulties. On the other hand, effective transmission is possible at high frequencies. So, modulation is always done in communication systems.

Key Point: The phenomenon of superposition of an information signal over a high frequency carrier wave is called modulation.

Previous Year Questions of Communication Systems

1. A TV transmission tower has a height of 140 m and the height of the receiving antenna is 40 m. What is the maximum distance upto which signals can be broadcasted from this tower in LOS (Line of Sight) mode? (Given: radius of earth = m)

- a. 65 km
- b. 48 km
- c. 40 km
- d. 80 km

Sol:

Given,

Height of transmission tower (h_T) = 140 m,

Height of receiving tower (h_R) = 40 m,

Radius of earth (R) = _____ m,

In this question, the height of the transmission tower and height of the receiving antenna both are given. So, we will apply the formula:

$$\frac{h_T + h_R}{R} = \frac{d^2}{4R^2}$$

So, Option - (a) is the correct answer.

Trick: If the height of receiving antenna is also given, then maximum line of sight distance (d_M),

2. A signal of frequency 20 kHz and peak voltage of 5 volts is used to modulate a carrier wave of frequency 1.2 MHz and peak voltage 25 volts. Choose the correct statement :

- a. Modulation index = 5, side frequency bands are at 1400 kHz and 1000 kHz**
- b. Modulation index = 0.2, side frequency bands are at 1220 kHz and 1180 kHz**
- c. Modulation index = 0.8, side frequency bands are at 1180 kHz and 1220 kHz**
- d. Modulation index = 5, side frequency bands are at 21.2 kHz and 18.8 kHz**

Sol:

Given,

Frequency of signal (f_m) = 20 kHz,

Peak voltage of signal (V_m) = 5 V,

Frequency of carrier wave (f_c) = 1.2 MHz = 1200 kHz,

Peak voltage of signal (V_m) = 25 V,

Now, Modulation index (μ) is given by,

—

Therefore, $\mu = \frac{1}{5} = 0.2$

Value of modulation index is 0.2

Side bands are given by:

$$= 1200 \text{ kHz} - 20 \text{ kHz} = 1180 \text{ kHz},$$

$$= 1200 \text{ kHz} + 20 \text{ kHz} = 1220 \text{ kHz},$$

Therefore, the right answer is option b.

Trick: Questions based on similar formulae will be asked in exams. So, be prepared for it.

Practice Questions

1. The maximum amplitude of an AM wave is found to be 15 V while its minimum amplitude is found to be 3V. What is the modulation index?

Ans: 0.667

2. A TV transmission tower antenna is at a height of 20 m. How much service area can it cover if the receiving antenna is (i) at ground level, (ii) at a height of 25 m? Calculate the percentage increase in area covered in case (ii) relative to case (i)

Ans: (i) 804 km² (ii) 3608 km² (iii) 349 %

Conclusion

In this article we have provided important information regarding the chapter Electromagnetic Induction such as important concepts, formulae, etc.. Students should work on more solved examples for securing good grades in the NEET exam.

Hi Aditya,

Best courses for you

**Full syllabus LIVE
courses**

Starting from ₹32,400

**One-to-one LIVE
classes**

Starting from ₹0/hr

Book your **Free Demo** session

Get a flavour of LIVE classes here at Vedantu

Book a free demo

Vedantu Improvement Promise

We promise improvement in
marks or get your fees
back. T&C Apply*